Technical Supplement: The Logic Hardening Layer (LHL)

Project Code: v3.5502
Subsystem: PD-Differential Decision Matrix

1. What is the Comparator?

In this architecture, the comparator is a high-speed analog-to-digital decision primitive. Unlike a multi-bit Analog-to-Digital Converter (ADC) that quantizes signals into complex, resource-heavy data packets, the comparator performs a singular physical decision. It acts as the "Event Horizon" where the continuous, overlapping Stokes vectors of the quartz medium are collapsed into discrete binary states (0 or 1).

2. Why Use a Comparator? (The Anti-Inversion Strategy)

The primary challenge in non-coherent polarization computing is Computational Inversion—a state where the energy and latency required for a digital processor to interpret a raw analog signal exceed the utility of the optical calculation itself.

- Zero-Latency Hardening: It eliminates the need for CPU-intensive sampling and Digital Signal Processing (DSP). The decision occurs at the speed of electron drift within the silicon, bypassing software instruction cycles.
- Noise Immunity (Hysteresis): By implementing a Schmitt-trigger configuration, the comparator ignores the micro-fluctuations and thermal noise inherent in non-coherent light environments, ensuring only significant polarization state shifts trigger a logical output.
- Protocol Alignment: Humans and digital systems cannot natively interact with a continuous wave. The comparator translates the "Nature's Rhythm" of the quartz into the "Human Logic" of the system bus.

3. Theory of Operation: Geometric Thresholding

The operation leverages a 5-Channel Differential Photo-Diode (PD) Array to perform spatial-to-logical mapping.

A. Input Vector Mapping

The PD array captures the Stokes parameters (S1, S2, S3) via differential intensity measurements:

$S1 = IH - IV$ (Horizontal vs. Vertical)
$S2 = I45 - I135$ (Diagonal vs. Anti-diagonal)
$S3 = IR - IL$ (Right vs. Left Circular)

These differential signals are fed directly into the comparator's inputs. To ensure robustness against source fluctuations, we utilize a Dynamic Reference (Vref) derived from the total intensity (S0).
The logic output is determined as:
Vout = High (1) if Vsig > Vref ± ΔH
Vout = Low (0) if Vsig < Vref ∓ ΔH
where ΔH represents the hysteresis window.

B. Unitary Operator Validation & Real-time Monitoring

The fundamental integrity of the optical computation is guaranteed by the Unitarity of the operator matrices implemented by the quartz crystal and the SLM (Spatial Light Modulator). In the EDA backend, all logic masks are validated against the unitary condition to ensure zero information loss during the evolution phase.

This unitarity is continuously verified in real time using the Si data from the PD array. Any significant drift in these values from the expected state—indicating physical displacement or thermal expansion—triggers an automated "Logic Realignment" procedure to recalibrate the optical path.

C. Physical State Collapse

As the polarization state rotates on the Poincaré Sphere due to the quartz crystal's geometric phase evolution, its projection on the measurement axes changes. When the vector crosses the "Equatorial Decision Plane" defined by the comparator's threshold, the hardware forces an instantaneous logical flip. The 5-channel differential design ensures these projections are captured with high contrast and stability, even in low-coherence conditions.

D. The Result: Causal-Chain Constraint (C^3)

The output is a clean, jitter-free square wave. This pulse represents a Causal-Chain Constraint (C^3)—a physical proof that the quartz logic has reached a specific state. This output is ready for immediate consumption by a general-purpose processor's GPIO or for triggering downstream SLM control loops without overhead.

Engineering Note:
The comparator is the Filter of Reality. By discarding the "process" of the quartz evolution and only reporting the "result," we achieve a high-fidelity interaction that remains robust even in non-coherent, high-entropy environments.

"Quartz calculates via evolution; the comparator decides via projection. Without the latter, the former remains silent to the human world."

Disclaimer:

The proposed non-coherent geometric polarization computing method is a pioneering exploration in an uncharted research field. Any errors or inaccuracies in the method are purely accidental, and no liability shall be assumed for any consequences arising therefrom.